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In [3]: import pandas as pd
import numpy as np
import re
import nltk
import string
import joblib

from sklearn.preprocessing import LabelEncoder
from sklearn.feature_extraction.text import TfidfVectorizer

from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
```

```
In [4]: # Step 2: Download NLTK data
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[4]: True

```
In [5]: # Step 3: Create sample data
data = {
    'text': [
        "Cats are running in the garden!",
        "The dog barked loudly at night.",
        "Birds are flying over the hills.",
        "She is reading a book on the balcony.",
    ],
    'label': ['animal', 'animal', 'animal', 'human']
}
df = pd.DataFrame(data)
print("Step Original Data:\n", df, "\n")
```

Step Original Data:

	text	label
0	Cats are running in the garden!	animal
1	The dog barked loudly at night.	animal
2	Birds are flying over the hills.	animal
3	She is reading a book on the balcony.	human

```
In [6]: # Step 4: Clean text
def clean_text(text):
    text = text.lower()
    text = re.sub(r'\d+', '', text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

```
df['clean_text'] = df['text'].apply(clean_text)
print("Step Cleaned Text:\n", df[['text', 'clean_text']], "\n")
```

Step Cleaned Text:

	text	clean_text
0	Cats are running in the garden!	cats are running in the garden
1	The dog barked loudly at night.	the dog barked loudly at night
2	Birds are flying over the hills.	birds are flying over the hills
3	She is reading a book on the balcony.	she is reading a book on the balcony

```
In [7]: # Save cleaned data
df[['text', 'clean_text']].to_csv('step4_cleaned_text.csv', index=False)
```

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In [8]: # Step 5: Lemmatization & Stop word removal
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

def preprocess_text(text):
    words = nltk.word_tokenize(text)
    lemmatized = [lemmatizer.lemmatize(word) for word in words if word not in stop_words]
    return ' '.join(lemmatized)

df['processed_text'] = df['clean_text'].apply(preprocess_text)
print("Step Processed Text (lemmatized & stopwords removed):\n", df[['clean_text', 'processed_text']], "\n")
```

Step Processed Text (lemmatized & stopwords removed):

	clean_text	processed_text
0	cats are running in the garden	cat running garden
1	the dog barked loudly at night	dog barked loudly night
2	birds are flying over the hills	bird flying hill
3	she is reading a book on the balcony	reading book balcony

```
In [9]: # Save processed data
df[['text', 'processed_text']].to_csv('step5_processed_text.csv', index=False)
```

```
In [10]: # Step 6: Label Encoding
label_encoder = LabelEncoder()
df['label_encoded'] = label_encoder.fit_transform(df['label'])
print("Step Encoded Labels:\n", df[['label', 'label_encoded']], "\n")
```

Step Encoded Labels:

	label	label_encoded
0	animal	0
1	animal	0
2	animal	0
3	human	1

```
In [11]: # Save Label encoder and encoded labels
joblib.dump(label_encoder, 'step6_label_encoder.pkl')
df[['label', 'label_encoded']].to_csv('step6_label_encoded.csv', index=False)
```

```
In [12]: # Step 7: TF-IDF Vectorization
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(df['processed_text'])
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=tfidf.get_feature_names_out())
print("Step TF-IDF Matrix (Top 5 rows):\n", tfidf_df.head(), "\n")
```

Step TF-IDF Matrix (Top 5 rows):

	balcony	barked	bird	book	cat	dog	flying	garden	hill	\
0	0.00000	0.0	0.00000	0.00000	0.57735	0.0	0.00000	0.57735	0.00000	
1	0.00000	0.5	0.00000	0.00000	0.00000	0.5	0.00000	0.00000	0.00000	
2	0.00000	0.0	0.57735	0.00000	0.00000	0.0	0.57735	0.00000	0.57735	
3	0.57735	0.0	0.00000	0.57735	0.00000	0.0	0.00000	0.00000	0.00000	

	loudly	night	reading	running
0	0.0	0.0	0.00000	0.57735
1	0.5	0.5	0.00000	0.00000
2	0.0	0.0	0.00000	0.00000
3	0.0	0.0	0.57735	0.00000

```
In [13]: # Save TF-IDF matrix and vectorizer
tfidf_df.to_csv('step7_tfidf_matrix.csv', index=False)
joblib.dump(tfidf, 'step7_tfidf_vectorizer.pkl')
```

```
Out[13]: ['step7_tfidf_vectorizer.pkl']
```

```
In [14]: # Final saved output summary
print(" All steps completed and outputs saved successfully!\n")
print(" Files Saved:")
print("- step4_cleaned_text.csv")
print("- step5_processed_text.csv")
print("- step6_label_encoded.csv")
print("- step6_label_encoder.pkl")
print("- step7_tfidf_matrix.csv")
print("- step7_tfidf_vectorizer.pkl")
```

All steps completed and outputs saved successfully!

Files Saved:

- step4_cleaned_text.csv
- step5_processed_text.csv
- step6_label_encoded.csv
- step6_label_encoder.pkl
- step7_tfidf_matrix.csv
- step7_tfidf_vectorizer.pkl

```
In [ ]:
```